NASIM MOSAYEBI

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PROFILE

Computer Science graduate (2:1, Brunel University London) applying for postgraduate study in Artificial Intelligence. My final-year project, Physio-AI, was a real-time physiotherapy feedback system combining computer vision and machine learning, awarded an A grade. Alongside my degree I develop mobile software independently: I contributed core game logic and SwiftUI interfaces to PathLife, published on the App Store, and am rebuilding it for Android in React Native, while building my own game, PawLife, from scratch. A prior BSc in Industrial Engineering and a Quality Assurance internship in medical device regulatory compliance inform my interest in applying AI safely within healthcare.

EDUCATION

BSc (Hons) Computer Science — 2:1 | *Brunel University London* *2023 – 2026*

- Final-year project graded A: AI-Based Physiotherapy Exercise Real-Time Feedback System.
- Relevant modules: Artificial Intelligence (machine learning, neural networks, deep learning), Software Engineering, Databases, Web Development.

Foundation in Computer Science | *Brunel University London* *2022 – 2023*

BSc Industrial Engineering | *Tehran Markaz Azad University, Iran* *Sep 2018 – 2022 (awarded 2024)*

- Completed all coursework in 2022 prior to relocating to the UK. Remaining final examinations were sat in 2024 during a return visit, and the degree was awarded that year.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS

Machine Learning: scikit-learn, MediaPipe Pose

Frontend & Mobile: React, React Native, Expo, Zustand, Swift, SwiftUI

Backend & Data: Node.js, Express, Spring Boot, REST APIs, SQLite, H2, WebSockets

Tools & Platforms: Git/GitHub, VS Code, Eclipse, WindowBuilder, Figma, Postman, Wireshark, Visual Paradigm, Railway, Cloudflare

Methodologies: Agile/Scrum, Waterfall, JUnit testing

KEY PROJECTS

Physio-AI — Final Year Project | *MediaPipe Pose, scikit-learn, Python* *Grade: A*

- Built an AI-powered physiotherapy system that analyses exercise technique in real time through a standard webcam, addressing the absence of feedback in unsupervised home rehabilitation.
- Used MediaPipe Pose to extract body landmarks from live video and converted them into pose embeddings for classification.
- Trained and compared multiple machine learning models; the best-performing Random Forest classifier (scikit-learn) achieved 96.23% accuracy, evaluated using confusion matrices and ROC analysis.
- Delivered a full real-time feedback pipeline: Python back end streaming to a JavaScript front end over WebSockets, with session data persisted in SQLite.
- Extended the system with fatigue detection and confidence-based visual feedback. Supervised by Dr David Bell, Department of Computer Science, Brunel University London.

PathLife — iOS Developer & Android Port | *React Native, TypeScript, Node.js* *Published on the App Store*

- Contributed core business logic and SwiftUI user interfaces to the published iOS version, with screens designed in Figma.
- Independently rebuilding the app in React Native and TypeScript for Android, translating all major screens, data models, state management and game systems.
- Designed, built and deployed the official support website (Node.js, Express, SQLite, Railway) with an admin dashboard, resolving an Apple App Store support-URL rejection.

App Store: apps.apple.com/gb/app/pathlife/id6769602618 · Support site: pathlifestudio.com

PawLife — Independent Mobile Game (in development) | *React Native, Expo*

- Designing and building a pet life-simulation game from scratch, including a custom illness system, minigames and a 25-achievement system; in final development ahead of release.
- Built a custom state-based navigation system, managed application state with Zustand, and implemented a shareable results card and persistent leaderboard using AsyncStorage.

ADDITIONAL ACADEMIC PROJECTS

CO2 Tracker — Level 2 Group Project | *React, Spring Boot, H2, Figma*

- Built the customer-facing React interface and a Spring Boot REST API (H2 database) for a sustainability web app tracking product carbon footprints via search, image upload and barcode lookup, aligned with UN SDG 12.
- Designed the interface in Figma from an initial hand-drawn sketch, then served as Scrum Master for Sprint 1, planning tasks and coordinating the team through GitHub and Trello.

Hospital Appointment System — Group Project | *Java, Swing, WindowBuilder, JUnit*

- Developed the appointment booking module — hospital search, department selection, and appointment scheduling, modification and cancellation — with CSV-based data persistence and JUnit test coverage.

EXPERIENCE

IT Administrator (Part-Time) | *ATOZ Education Services, London, United Kingdom* *Apr 2024 – Jun 2026*

- Provided IT support for hardware, software and network issues across the organisation.
- Maintained IT systems and ensured reliable day-to-day operations for staff and students.
- Assisted staff with technical troubleshooting and system setup, communicating solutions to non-technical users.

Quality Assurance Intern, Regulatory Compliance | *Farsar Tejarat, Tehran, Iran* *Jan 2021 – Jan 2022*

- Assisted with process validation activities supporting regulatory compliance requirements across three to five medical device product lines.
- Participated in FMEA and risk assessment activities, and assisted in internal and external quality audits by preparing documentation and traceability records.
- Supported CAPA activities by tracking corrective actions and documenting quality improvements.
- Collaborated with a QA team of six to ten across Quality Assurance, Production and R&D to support product quality before release.

IT Support Assistant | *Imam Hossein Hospital, Karaj, Iran* *Apr 2020 – Sep 2020*

- Installed and configured computer hardware, software and network systems within a hospital IT environment, keeping essential systems running for clinical and administrative staff.
- Diagnosed and resolved hardware and software faults to minimise downtime, collaborating with hospital staff on complex issues.

IT Support Assistant | *Nedaye Salamat Farsar Alborz, Karaj, Iran* *Apr 2019 – Sep 2019*

- Provided first-line technical support for CRM users at a company specialising in equipping and managing hospital and healthcare facilities.
- Assisted employees with CRM usage and troubleshooting, and delivered onboarding and basic CRM training for new staff.
- Managed user accounts and access permissions, keeping account data accurate and secure.

LANGUAGES

English (Fluent) · Persian (Native) · German (B1) · Arabic (A1)